Probability and Statistics in Full

Extended mathematics notes Sources: listed under References

What this file is really about:

$$\text{counting outcomes} \rightarrow \text{distributions} \rightarrow \text{estimates with honest uncertainty} \rightarrow \text{information}$$

How to use this file. This is a reference that starts from the axioms. Each result here is used elsewhere in the course. Part B checks the results by simulation.

Step-up path

You need. Fractions, sums and basic calculus.

1. Use total probability.
2. Compute a mean and variance.
3. Pick a distribution.
4. Build a confidence interval.
5. Compute entropy and KL.

You can stop when you can derive any loss from a likelihood.

Contents

Part A Maths From the Ground Up

A1 Sets, sample spaces and the axioms

The **sample space** Ω is the set of possible outcomes. An **event** is a subset of it. Probability obeys three **axioms**:

1. $P(A) \geq 0$;
2. $P(\Omega) = 1$;
3. for disjoint events, $P(A \cup B) = P(A) + P(B)$.

Everything else follows from these. Two results are used constantly: $P(A^c) = 1 - P(A)$ and $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. The **union bound** follows: $P(\bigcup_i A_i) \leq \sum_i P(A_i)$.

A2 Counting

A **factorial** counts orderings of n items: $n!$. The number of ordered selections of k is $n!/(n-k)!$. The **binomial coefficient** counts unordered selections:

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}.$$

A3 Conditional probability and independence

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

A and B are **independent** if $P(A \cap B) = P(A)P(B)$. The **law of total probability** splits an event over a partition B_i:

$$P(A) = \sum_i P(A \mid B_i)\, P(B_i).$$

Bayes' rule combines these: $P(B_i \mid A) = P(A \mid B_i)P(B_i)/P(A)$.

A4 Random variables, mass, density and CDF

A **random variable** assigns a number to each outcome. A discrete one has a **probability mass function** $p(x) = P(X = x)$. A continuous one has a **density** $p(x)$, where $P(a < X < b) = \int_a^b p$. A density can exceed 1, but it must integrate to 1. The **cumulative distribution function** is $F(x) = P(X \le x)$. It rises from 0 to 1, and $F' = p$. If U is uniform, then $F^{-1}(U)$ has CDF F, which is inverse-CDF sampling.

A **joint distribution** $p(x, y)$ gives both at once. The **marginal** sums out the other variable, $p(x) = \int p(x, y)\, dy$.

A5 Expectation

$\mathbb{E}[X] = \sum_x x p(x)$, or $\int x p(x)\, dx$. Three properties do most of the work.

- **Linearity of expectation**: $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$, whether or not they are independent.
- **Law of the unconscious statistician**: $\mathbb{E}[g(X)] = \int g(x)p(x)\, dx$, with no need for the density of $g(X)$.
- **Law of total expectation**, or the tower property: $\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]]$.

A6 Variance, covariance and correlation

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2] - \mu^2, \qquad \mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

The rules are $\mathrm{Var}(aX + b) = a^2\mathrm{Var}(X)$ and $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\mathrm{Cov}(X, Y)$. **Correlation** divides by both standard deviations and lies in $[-1, 1]$. Independence implies zero covariance, but zero covariance does not imply independence: take $Y = X^2$ with symmetric X.

The **law of total variance** splits variance into explained and unexplained parts:

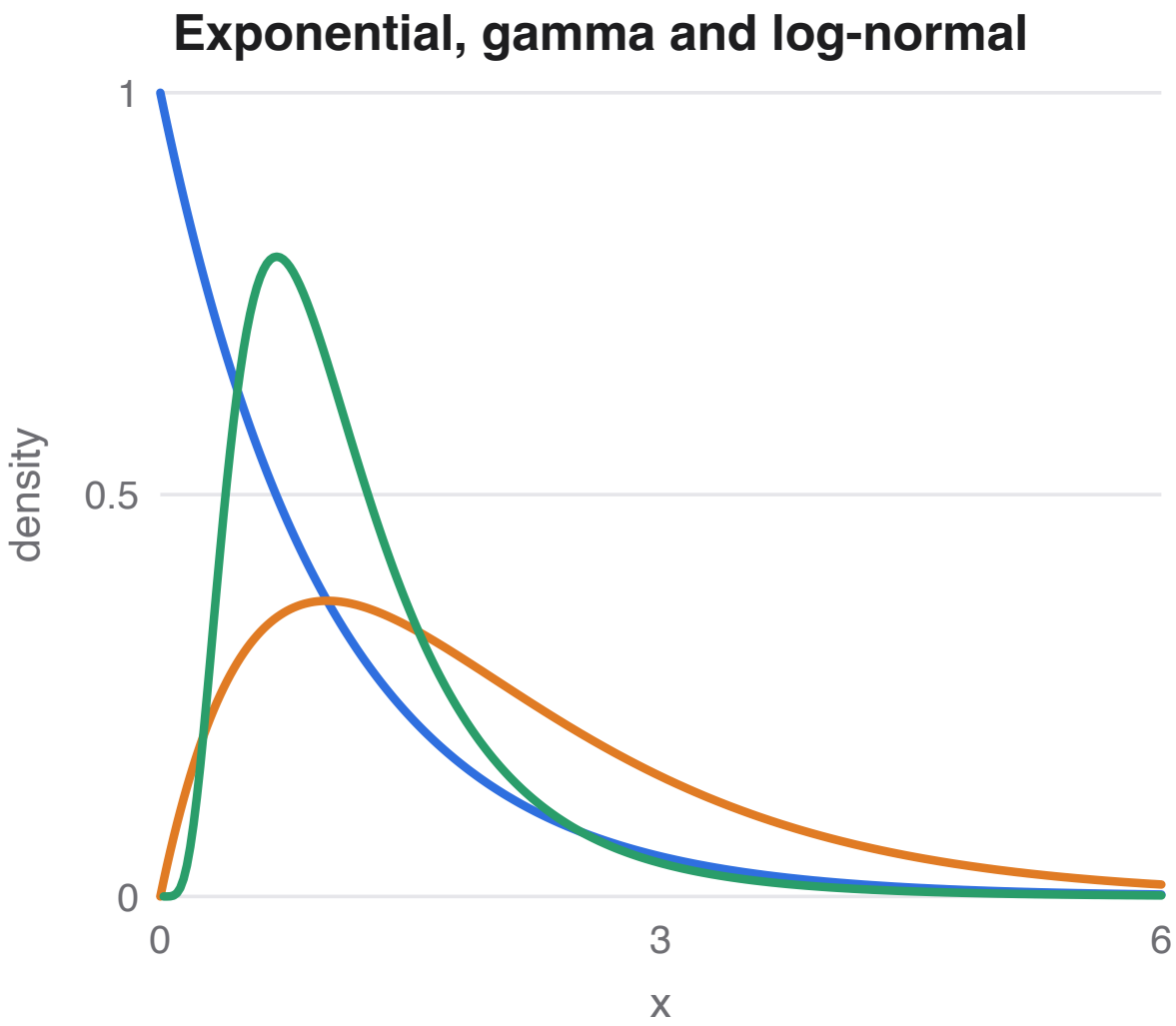
$$\mathrm{Var}(X) = \mathbb{E}[\mathrm{Var}(X \mid Y)] + \mathrm{Var}(\mathbb{E}[X \mid Y]).$$

A7 Discrete distributions

Name	$p(x)$	Mean	Variance	Models
Bernoulli(p)	$p^x(1-p)^{1-x}$	p	$p(1-p)$	One binary label
Binomial(n, p)	$\binom{n}{x}p^x(1-p)^{n-x}$	np	$np(1-p)$	Successes in n trials
Categorical	$\prod_k p_k^{[x=k]}$			One class label
Multinomial	$\frac{n!}{\prod x_k!}\prod p_k^{x_k}$	np_k	$np_k(1-p_k)$	Word counts
Geometric(p)	$(1-p)^{x-1}p$	$1/p$	$(1-p)/p^2$	Trials until a success
Poisson(λ)	$e^{-\lambda}\lambda^x/x!$	λ	λ	Counts per interval

A8 Continuous distributions

Name	Density	Mean	Variance	Models
Uniform(a, b)	$1/(b-a)$	$(a+b)/2$	$(b-a)^2/12$	No preference
Exponential distribution(λ)	$\lambda e^{-\lambda x}$	$1/\lambda$	$1/\lambda^2$	Waiting time
Normal(μ, σ^2)	$\frac{1}{\sigma\sqrt{2\pi}}e^{-(x-\mu)^2/2\sigma^2}$	μ	σ^2	Sums of many effects
Laplace distribution(μ, b)	$\frac1{2b}e^{-\|x-\mu\|/b}$	μ		
Gamma distribution(α, β)	$\frac{\beta^\alpha}{\Gamma(\alpha)}x^{\alpha-1}e^{-\beta x}$	α/β	α/β^2	Positive quantities, rate priors
Beta distribution(a, b)	$\frac{x^{a-1}(1-x)^{b-1}}{B(a,b)}$	$\frac{a}{a+b}$	$\frac{ab}{(a+b)^2(a+b+1)}$	A probability
Log-normal	$\ln X$ is normal	$e^{\mu+\sigma^2/2}$		Positive skewed values
Student's t(ν)	Heavy tails	0 for $\nu > 1$	$\nu/(\nu-2)$ for $\nu > 2$	Means with unknown variance
Chi-squared(k)	Sum of k squared normals	k	$2k$	Variances and goodness of fit
Dirichlet(α)	$\propto \prod_k x_k^{\alpha_k-1}$	$\alpha_k/\sum\alpha$		A probability vector



A9 Conjugate priors and the exponential family

A prior is **conjugate** if the posterior has the same form.

Likelihood	Prior	Posterior
Bernoulli	Beta(a, b)	Beta$(a + s,\ b + f)$
Categorical	Dirichlet(α)	Dirichlet$(\alpha + \text{counts})$
Poisson	Gamma(α, β)	Gamma$(\alpha + \sum x,\ \beta + n)$
Normal with known variance	Normal	Normal, combining precisions

The **exponential family** has the form $p(x \mid \eta) = h(x)\exp(\eta^\top T(x) - A(\eta))$. Bernoulli, Poisson, normal and many others belong. $\nabla A(\eta) = \mathbb{E}[T(x)]$, and T is a **sufficient statistic**: once you know it, the rest of the data adds nothing about η. Generalised linear models are built on this family.

A10 Moment generating functions

$M(t) = \mathbb{E}[e^{tX}]$ produces moments by differentiation, $M^{(k)}(0) = \mathbb{E}[X^k]$. For independent variables, $M_{X+Y} = M_X M_Y$. This is the standard tool for proving the central limit theorem and exponential tail bounds.

A11 Inequalities

Markov's inequality: for $X \geq 0$, $P(X \geq a) \leq \mathbb{E}[X]/a$.

Chebyshev's inequality: $P(|X - \mu| \geq k\sigma) \leq 1/k^2$.

Hoeffding's inequality: for independent variables in $[0, 1]$ with sample mean $\bar{X}$,

$$P(|\bar{X} - \mu| \geq t) \leq 2e^{-2nt^2}.$$

Tail bounds of this kind are the basis of generalisation theory, confidence bounds for bandits and sample size calculations.

A12 Law of large numbers and central limit theorem

The **law of large numbers**: the mean of n independent copies converges to μ.

The **central limit theorem**: for finite variance,

$$\sqrt{n}\,(\bar{X} - \mu)/\sigma \to \mathcal{N}(0, 1).$$

This is why errors of averages are often normal, and why the **standard error** is $\sigma/\sqrt{n}$.

A13 Order statistics

Sort a sample as $X_{(1)} \leq \cdots \leq X_{(n)}$. The maximum of n independent copies has CDF $F(x)^n$. The minimum has $1 - (1 - F(x))^n$. This is why, in high dimensions, nearest and farthest neighbours come to have similar distances, and why the largest of many noisy estimates is biased upwards.

A14 Estimators

An **estimator** is a function of the sample. Its bias is $\mathbb{E}[\hat{\theta}] - \theta$. It is **unbiased** if that is zero, and **consistent** if it converges to θ as n grows. Mean squared error decomposes as $\text{bias}^2 + \text{variance}$.

The sample variance divides by $n - 1$, because using $\bar{x}$ in place of μ removes one degree of freedom.

Maximum likelihood picks $\arg\max_\theta \sum_i \ln p(x_i \mid \theta)$. It is consistent under regular conditions and its asymptotic variance is the inverse Fisher information. **MAP** adds a log prior.

A15 Confidence intervals and bootstrap

A 95% **confidence interval** procedure captures the true value in 95% of repeated samples. It is not a probability that this particular interval contains the value. For a mean it is $\bar{x} \pm t \cdot s/\sqrt{n}$.

The **bootstrap** resamples the data with replacement and recomputes the statistic. It gives intervals for quantities with no simple formula.

Kernel density estimation estimates a density by placing a small bump at each point:

$$\hat{p}(x) = \frac{1}{nh} \sum_i K\left(\frac{x - x_i}{h}\right).$$

The bandwidth h trades bias against variance.

A16 Hypothesis tests

A test sets a null hypothesis, computes a statistic and reports a **p-value**: the probability under the null of a result at least as extreme. It is not the probability that the null is true.

Error	Meaning	Rate
Type I	Rejecting a true null	α
Type II	Missing a real effect	β

The **power of a test** is $1 - \beta$. It rises with effect size and sample size.

- **t-test**: compare means, $t = (\bar{x} - \mu_0)/(s/\sqrt{n})$.

- **Chi-squared test**: compare counts, $\sum (O-E)^2/E$.
- **ANOVA**: compare several means, using between-group variance over within-group variance.

Multiple testing. Running m tests at level α inflates false alarms. The **Bonferroni** correction tests each at α/m.

A17 Stochastic processes and Markov chains

A **stochastic process** is a family of random variables indexed by time. A **Markov chain** has a future that depends only on the present. With **transition matrix** $P_{ij} = P(x_{t+1} = j \mid x_t = i)$, a distribution evolves by $\pi_{t+1} = \pi_t P$. A **stationary distribution** satisfies $\pi = \pi P$.

A **martingale** satisfies $\mathbb{E}[X_{t+1} \mid \text{history}] = X_t$, a fair game with no drift. It is the basis of concentration results for dependent data.

A18 Information theory

Quantity	Definition
Entropy	$H(X) = -\sum_x p \log p$
Joint entropy	$H(X,Y) = -\sum p(x,y) \log p(x,y)$
Conditional entropy	$H(Y \mid X) = H(X,Y) - H(X)$
Mutual information	$I(X;Y) = H(Y) - H(Y \mid X)$
Cross-entropy	$H(p,q) = -\sum p \log q$
Kullback-Leibler divergence	$\mathrm{KL}(p\|q) = \sum p \log(p/q)$
Jensen-Shannon	$\frac{1}{2}\mathrm{KL}(p\|m) + \frac{1}{2}\mathrm{KL}(q\|m)$, with $m = \frac{1}{2}(p+q)$

Logs in base 2 give bits, and natural logs give nats.

Gibbs' inequality states $\mathrm{KL}(p\|q) \geq 0$, with equality only when $p = q$. It follows from Jensen. So cross-entropy equals $H(p) + \mathrm{KL}(p\|q)$, and minimising it over q matches the distribution. KL is not symmetric.

The **data processing inequality**: if $X \to Y \to Z$ is a Markov chain, then $I(X;Z) \leq I(X;Y)$. Processing cannot add information about X.

Between two normals with the same dimension d,

$$\mathrm{KL}(\mathcal{N}_0 \| \mathcal{N}_1) = \tfrac{1}{2}\Big[\mathrm{Tr}(\Sigma_1^{-1}\Sigma_0) + (\mu_1 - \mu_0)^\top \Sigma_1^{-1}(\mu_1 - \mu_0) - d + \ln \frac{\det \Sigma_1}{\det \Sigma_0}\Big].$$

Part B In Code

B1 Total probability, linearity and total variance

```
import numpy as np
rng = np.random.default_rng(0)
n = 2_000_000
g = rng.integers(0, 2, n); y = rng.normal(np.where(g, 3.0, 0.0), np.where(g, 2.0, 1.0))
within = 0.5 * 4 + 0.5 * 1; between = 0.25 * 9     # variance of the conditional means 0 ar
print(round(y.var(), 3), within + between)
assert abs(y.var() - (within + between)) < 0.02
x = rng.normal(size=n); z = x**2
assert abs(np.mean(x * z)) < 0.01        # zero covariance, yet z depends on x
assert abs(np.mean(x + 3 * z) - 3) < 0.01
```

B2 Distribution summaries

```
import numpy as np
rng = np.random.default_rng(1); n = 3_000_000
checks = [('geometric', rng.geometric(0.25, n), 4, 12), ('poisson', rng.poisson(3.5, n), 3.
          ('exponential', rng.exponential(2.0, n), 2, 4), ('gamma', rng.gamma(3, 0.5, n), 1
          ('beta', rng.beta(2, 5, n), 2/7, 10/(49*8)), ('laplace', rng.laplace(0, 2, n), 0,
          ('chi-squared', rng.chisquare(4, n), 4, 8)]
for name, s, m, v in checks:
    assert abs(s.mean() - m) < 0.02 and abs(s.var() - v) < 0.05 * max(v, 1), name
s = rng.lognormal(0.2, 0.5, n); assert abs(s.mean() - np.exp(0.2 + 0.125)) < 0.01
print('all distribution formulas agree')
```

B3 Tail bounds really are bounds

```
import numpy as np
rng = np.random.default_rng(2)
x = rng.exponential(1.0, 3_000_000)
for a in (2, 4): assert np.mean(x >= a) <= 1 / a
for k in (2, 3): assert np.mean(np.abs(x - 1) >= k) <= 1 / k**2
n, t = 100, 0.1
means = rng.binomial(n, 0.5, 2_000_000) / n
actual = np.mean(np.abs(means - 0.5) >= t); bound = 2 * np.exp(-2 * n * t * t)
print(round(actual, 4), round(bound, 4)); assert actual <= bound
```

B4 The central limit theorem and the maximum

```
import numpy as np
rng = np.random.default_rng(3)
u = rng.random((400_000, 30)); z = (u.mean(1) - 0.5) / (np.sqrt(1/12) / np.sqrt(30))
assert abs(np.mean(np.abs(z) < 1.96) - 0.95) < 0.003
m = u.max(1); assert abs(np.mean(m < 0.9) - 0.9**30) < 0.002
print('CLT and order statistics agree')
```

B5 A confidence interval has the coverage it claims

```
import numpy as np
from scipy import stats
rng = np.random.default_rng(4)
n = 12; R = 40000; t = stats.t.ppf(0.975, n - 1)
x = rng.normal(5, 2, (R, n))
m, s = x.mean(1), x.std(1, ddof=1)
cover = np.mean(np.abs(m - 5) <= t * s / np.sqrt(n))
print(round(cover, 4)); assert abs(cover - 0.95) < 0.006
p = rng.random((R, 20)); assert abs(np.mean(p.min(1) < 0.05) - (1 - 0.95**20)) < 0.01
assert np.mean(p.min(1) < 0.05 / 20) < 0.06
```

B6 Information identities

```
import numpy as np
P = np.array([[0.3, 0.1], [0.2, 0.4]])
h = lambda p: -np.sum(p[p > 0] * np.log2(p[p > 0]))
Hxy, Hx, Hy = h(P.ravel()), h(P.sum(1)), h(P.sum(0))
Hy_x = Hxy - Hx; I = Hy - Hy_x
assert I >= 0 and np.isclose(I, np.sum(P * np.log2(P / np.outer(P.sum(1), P.sum(0)))))
p, q = np.array([0.5, 0.3, 0.2]), np.array([0.2, 0.2, 0.6])
kl = np.sum(p * np.log2(p / q)); assert kl > 0 and np.isclose(-np.sum(p * np.log2(q)), h(p)
assert not np.isclose(kl, np.sum(q * np.log2(q / p)))
print(round(I, 4), round(kl, 4))
```

Part C Practice

P1. A 1% disease has a test with 90% sensitivity and a 5% false-positive rate. What is $P(\text{positive})$?

$0.01 \times 0.9 + 0.99 \times 0.05 = 0.0585$.

P2. What is the mean number of die rolls until the first six?

Geometric with $p = 1/6$, so 6.

P3. A Beta(2, 2) prior sees 3 heads and 1 tail. What is the posterior mean?

Beta(5, 3), with mean $5/8$.

P4. A non-negative variable has mean 2. Bound $P(X \geq 10)$.

By Markov, at most 0.2.

P5. What is the standard error of a mean with $\sigma = 10$ and $n = 100$?

1.

P6. Twenty tests are run at $\alpha = 0.05$, and every null is true. What is the chance of at least one false alarm if they are independent?

$1 - 0.95^{20} \approx 0.64$.

P7. Does zero mutual information mean independence?

Yes. $I(X;Y) = \mathrm{KL}(p(x,y) \| p(x)p(y))$ is zero only when they are equal.